Microprocessor 8085 - [Code Editor]

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Microprocessor 8085 Simulator Software Kit 1.0

Ln 6, Col 6

TStates : 79 | MCycles : 24 | Time of Execution : 79 µs

Flags : S Z AC P CY

General Regs.:

A	B	C	D	E	H	L	M	SP	PC
00	10	00	00	00	00	00	3A	0000	0000

```
LDA 4200h
MOV B A
LDA 4201h
ADD B
JNC SKIP
INR C
SKIP: STA 4202h
MOV A C
STA 4203h
HLT
```

Starting Address : 0000

Comments :

User Data Grid

Address	Data
4200	10
4201	05
4202	15
4203	00
0000	3A

Hex Code Grid

Address	Data
0000	3A
0001	00
0002	42
0003	47
0004	3A
0005	01
0006	42
0007	80
0008	D2
0009	0C
000A	00
000B	0C
000C	32
000D	02
000E	42
000F	79
0010	32
0011	03

1.Data Transfer Group :

MOV	MVI	LDA	STA
LHLD	SHLD	LXI	XCHG
LDAX	STAX	IN	OUT

2.Arithmatic Group :

ADD	ADI	ADC	ACI	DAA
SUB	SUI	SBB	SBI	DAD
INR	DCR	INX	DCX	

3.Logical Group :

ANA	ANI	XRA	XRI	CMA
ORA	ORI	CMP	CPI	CMC
RLC	RRC	RAL	RAR	STC

4.Branch and Machine Control Group

JMP	J_	CALL	C_	
RET	R_	EI	DI	PUSH
PCHL	RST	RIM	SIM	POP
XTHL	SPHL	NOP	HLT	

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Instruction Information :

MOVe -- copy from source to destination

Ln 12, Col 4

TStates : 79 | MCycles : 21 | Time of Execution : 79 µs

Flags : S Z AC P CY

General Regs.:	A	B	C	D	E	H	L	M	SP	PC
	18	00	01	00	00	42	03	01	0000	8000

```
LXI H 4200h
MVI C 00h
MOV A M
INX H
ADD M
JNC SKIP
INR C
SKIP: INX H
MOV M A
INX H
MOV M C
HLT
```

Starting Address : 8000

Comments :

User Data Grid

Address	Data
4200	2F
4201	E9
4202	18
4203	01

Hex Code Grid

Address	Data
8000	21
8001	00
8002	42
8003	0E
8004	00
8005	7E
8006	23
8007	86
8008	D2
8009	0C
800A	80
800B	0C
800C	23
800D	77
800E	23
800F	71
8010	76

1.Data Transfer Group :

MOV	MVI	LDA	STA
LHLD	SHLD	LXI	XCHG
LDAX	STAX	IN	OUT

2.Arithmatic Group :

ADD	ADI	ADC	ACI	DAA
SUB	SUI	SBB	SBI	DAD
INR	DCR	INX	DCX	

3.Logical Group :

ANA	ANI	XRA	XRI	CMA
ORA	ORI	CMP	CPI	CMC
RLC	RRC	RAL	RAR	STC

4.Branch and Machine Control Group

JMP	J_	CALL	C_	
RET	R_	EI	DI	PUSH
PCHL	RST	RIM	SIM	POP
XTHL	SPHL	NOP	HLT	

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Instruction Information :

LoaD Accumulator indirect

Ln 6, Col 4

TStates : 52 | MCycles : 16 | Time of Execution : 52 µs
Flags : S Z AC P CY

General Regs.:	A	B	C	D	E	H	L	M	SP	PC
	8B	4A	00	00	00	00	00	3A	0000	0000

```
LDA 4201h
MOV B A
LDA 4200h
SUB B
STA 4202h
HLT
```

Starting Address : 0000

Comments :

User Data Grid

Address	Data
4200	D5
4201	4A
4202	8B
0000	3A
0000	3A

Hex Code Grid

Address	Data
0000	3A
0001	01
0002	42
0003	47
0004	3A
0005	00
0006	42
0007	90
0008	32
0009	02
000A	42
000B	76

1.Data Transfer Group :

MOV	MVI	LDA	STA
LHLD	SHLD	LXI	XCHG
LDAX	STAX	IN	OUT

2.Arithmatic Group :

ADD	ADI	ADC	ACI	DAA
SUB	SUI	SBB	SBI	DAD
INR	DCR	INX	DCX	

3.Logical Group :

ANA	ANI	XRA	XRI	CMA
ORA	ORI	CMP	CPI	CMC
RLC	RRC	RAL	RAR	STC

4.Branch and Machine Control Group

JMP	J_	CALL	C_	
RET	R_	EI	DI	PUSH
PCHL	RST	RIM	SIM	POP
XTHL	SPHL	NOP	HLT	

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Instruction Information :

INput data to accumulator from a port with 8-bit address

Ln 10, Col 8

TStates : 88 | MCycles : 26 | Time of Execution : 88 µs

Flags : S Z AC P CY

General Regs.:	A	B	C	D	E	H	L	M	SP	PC
	01	DD	01	00	00	00	00	3A	0000	0000

```
LDA 4201h
MOV B A
LDA 4200h
SUB B
JNC SKIP
INR C
CMA
INR A
SKIP: STA 4202h
MOV A C
STA 4203h
HLT
```

Starting Address : 0000

Comments :

User Data Grid

Address	Data
4200	CF
4201	DD
4202	0E
4203	01
0000	3A

Hex Code Grid

Address	Data
0000	3A
0001	01
0002	42
0003	47
0004	3A
0005	00
0006	42
0007	90
0008	D2
0009	0E
000A	00
000B	0C
000C	2F
000D	3C
000E	32
000F	02
0010	42
0011	79

1.Data Transfer Group :

MOV	MVI	LDA	STA
LHLD	SHLD	LXI	XCHG
LDAX	STAX	IN	OUT

2.Arithmatic Group :

ADD	ADI	ADC	ACI	DAA
SUB	SUI	SBB	SBI	DAD
INR	DCR	INX	DCX	

3.Logical Group :

ANA	ANI	XRA	XRI	CMA
ORA	ORI	CMP	CPI	CMC
RLC	RRC	RAL	RAR	STC

4.Branch and Machine Control Group

JMP	J_	CALL	C_	
RET	R_	EI	DI	PUSH
PCHL	RST	RIM	SIM	POP
XTHL	SPHL	NOP	HLT	

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Instruction Information :

INput data to accumulator from a port with 8-bit address

Ln 11, Col 4

TStates : 6165 | MCycles : 1799 | Time of Execution : 6165 µs

Flags : S Z AC P CY

General Regs.:

A	B	C	D	E	H	L	M	SP	PC
FE	00	00	00	6D	6C	26	00	0000	8000

```
LDA 4200h
MOV E A
LDA 4201h
MOV B A
LXI H 0000h
MVI D 00h
NEXT: DAD D
DCR B
JNZ NEXT
SHLD 4202h
HLT
```

Starting Address : 8000

Comments :

User Data Grid

Address	Data
4200	6D
4201	FE
4202	26
4203	6C

Hex Code Grid

Address	Data
8000	3A
8001	00
8002	42
8003	5F
8004	3A
8005	01
8006	42
8007	47
8008	21
8009	00
800A	00
800B	16
800C	00
800D	19
800E	05
800F	C2
8010	0D
8011	80

1.Data Transfer Group :

MOV	MVI	LDA	STA
LHLD	SHLD	LXI	XCHG
LDAX	STAX	IN	OUT

2.Arithmatic Group :

ADD	ADI	ADC	ACI	DAA
SUB	SUI	SBB	SBI	DAD
INR	DCR	INX	DCX	

3.Logical Group :

ANA	ANI	XRA	XRI	CMA
ORA	ORI	CMP	CPI	CMC
RLC	RRC	RAL	RAR	STC

4.Branch and Machine Control Group

JMP	J_	CALL	C_	
RET	R_	EI	DI	PUSH
PCHL	RST	RIM	SIM	POP
XTHL	SPHL	NOP	HLT	

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Instruction Information :

Code Pad

Ln 14, Col 1

TStates : 521 | MCycles : 146 | Time of Execution : 521 µs

Flags : S Z AC P CY

General Regs.:

A	B	C	D	E	H	L	M	SP	PC
0F	0A	0F	00	00	00	00	3A	0000	0000

```
LDA 4201h
MOV B A
LDA 4200h
MVI C 00h
AGAIN: CMP B
JC STORE
SUB B
INR C
JMP AGAIN
STORE: STA 4203h
MOV A C
STA 4202h
HLT
```

Starting Address : 0000

Comments :

User Data Grid

Address	Data
4200	9F
4201	0A
4202	0F
4203	09
0000	3A

Hex Code Grid

Address	Data
0000	3A
0001	01
0002	42
0003	47
0004	3A
0005	00
0006	42
0007	0E
0008	00
0009	B8
000A	DA
000B	12
000C	00
000D	90
000E	0C
000F	C3
0010	09
0011	00

1.Data Transfer Group :

MOV	MVI	LDA	STA
LHLD	SHLD	LXI	XCHG
LDAX	STAX	IN	OUT

2.Arithmatic Group :

ADD	ADI	ADC	ACI	DAA
SUB	SUI	SBB	SBI	DAD
INR	DCR	INX	DCX	

3.Logical Group :

ANA	ANI	XRA	XRI	CMA
ORA	ORI	CMP	CPI	CMC
RLC	RRC	RAL	RAR	STC

4.Branch and Machine Control Group

JMP	J_	CALL	C_	
RET	R_	EI	DI	PUSH
PCHL	RST	RIM	SIM	POP
XTHL	SPHL	NOP	HLT	

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Instruction Information :

Rotate accumulator Left with Carry

Ln 21, Col 4

TStates : 2285 | MCycles : 604 | Time of Execution : 2285 µs

Flags : S Z AC P CY

General Regs.:

A	B	C	D	E	H	L	M	SP	PC
AB	00	00	34	00	42	07	F2	0000	8000

```
LDA 4200h
MOV B A
DCR B
LOOP2: LXI H 4200h
MOV C M
DCR C
INX H
LOOP1: MOV A M
INX H
CMP M
JC AHEAD
MOV D M
MOV M A
DCX H
MOV M D
INX H
AHEAD: DCR C
JNZ LOOP1
DCR B
JNZ LOOP2
HLT
```

Starting Address : 8000

Comments :

User Data Grid

Address	Data
4200	07
4201	34
4202	4F
4203	69
4204	84
4205	92
4206	AB
4207	F2

Hex Code Grid

Address	Data
8000	3A
8001	00
8002	42
8003	47
8004	05
8005	21
8006	00
8007	42
8008	4E
8009	0D
800A	23
800B	7E
800C	23
800D	BE
800E	DA
800F	16
8010	80
8011	56

1.Data Transfer Group :

MOV	MVI	LDA	STA
LHLD	SHLD	LXI	XCHG
LDAX	STAX	IN	OUT

2.Arithmatic Group :

ADD	ADI	ADC	ACI	DAA
SUB	SUI	SBB	SBI	DAD
INR	DCR	INX	DCX	

3.Logical Group :

ANA	ANI	XRA	XRI	CMA
ORA	ORI	CMP	CPI	CMC
RLC	RRC	RAL	RAR	STC

4.Branch and Machine Control Group

JMP	J_	CALL	C_	
RET	R_	EI	DI	PUSH
PCHL	RST	RIM	SIM	POP
XTHL	SPHL	NOP	HLT	

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Instruction Information :

LoaD Accumulator indirect



TStates : 2015 | MCycles : 541 | Time of Execution : 2015 µs

Flags : S Z AC P CY

General Regs.:

A	B	C	D	E	H	L	M	SP	PC
4F	00	00	F2	00	42	07	34	0000	8000

X

```
LDA 4200h
MOV B A
DCR B
LOOP2: LXI H 4200h
MOV C M
DCR C
INX H
LOOP1: MOV A M
INX H
CMP M
JNC AHEAD
MOV D M
MOV M A
DCX H
MOV M D
INX H
AHEAD: DCR C
JNZ LOOP1
DCR B
JNZ LOOP2
HLT
```

Starting Address : 8000

Comments :

User Data Grid

Address	Data
4200	07
4201	F2
4202	AB
4203	92
4204	84
4205	69
4206	4F
4207	34

Hex Code Grid

Address	Data
8000	3A
8001	00
8002	42
8003	47
8004	05
8005	21
8006	00
8007	42
8008	4E
8009	0D
800A	23
800B	7E
800C	23
800D	BE
800E	D2
800F	16
8010	80
8011	56

1.Data Transfer Group :

MOV	MVI	LDA	STA
LHLD	SHLD	LXI	XCHG
LDAX	STAX	IN	OUT

2.Arithmatic Group :

ADD	ADI	ADC	ACI	DAA
SUB	SUI	SBB	SBI	DAD
INR	DCR	INX	DCX	

3.Logical Group :

ANA	ANI	XRA	XRI	CMA
ORA	ORI	CMP	CPI	CMC
RLC	RRC	RAL	RAR	STC

4.Branch and Machine Control Group

JMP	J_	CALL	C_	
RET	R_	EI	DI	PUSH
PCHL	RST	RIM	SIM	POP
XTHL	SPHL	NOP	HLT	

☑ Auto Insert Insert Cancel

Instruction Information :

LoaD Accumulator indirect

Ln 7, Col 4

TStates : 59 | MCycles : 18 | Time of Execution : 59 µs

Flags : S Z AC P CY

General Regs.:

A	B	C	D	E	H	L	M	SP	PC
09	00	00	00	00	00	00	3A	0000	0000

```
LDA 4200h
SUI 30h
CPI 0Ah
JC STORE
SUI 07h
STORE: STA 4201h
HLT
```

Starting Address : 0000

Comments :

User Data Grid

Address	Data
4200	40
4201	09
0000	00
0000	00
0000	00
0000	00
0000	00
0000	00

Hex Code Grid

Address	Data
0000	3A
0001	00
0002	42
0003	D6
0004	30
0005	FE
0006	0A
0007	DA
0008	0C
0009	00
000A	D6
000B	07
000C	32
000D	01
000E	42
000F	76

1.Data Transfer Group :

MOV	MVI	LDA	STA
LHLD	SHLD	LXI	XCHG
LDAX	STAX	IN	OUT

2.Arithmatic Group :

ADD	ADI	ADC	ACI	DAA
SUB	SUI	SBB	SBI	DAD
INR	DCR	INX	DCX	

3.Logical Group :

ANA	ANI	XRA	XRI	CMA
ORA	ORI	CMP	CPI	CMC
RLC	RRC	RAL	RAR	STC

4.Branch and Machine Control Group

JMP	J_	CALL	C_	
RET	R_	EI	DI	PUSH
PCHL	RST	RIM	SIM	POP
XTHL	SPHL	NOP	HLT	

Auto Insert | Insert | Cancel

Instruction Information :

LoaD H and L register direct

Ln 10, Col 4

TStates : 309 | MCycles : 79 | Time of Execution : 309 µs

Flags : S Z AC P CY

General Regs.:

A	B	C	D	E	H	L	M	SP	PC
DD	42	07	43	07	00	00	01	0000	0000

```
LXI B 4200h
LXI D 4300h
MVI H 07h
NEXT: LDAX B
STAX D
INX B
INX D
DCR H
JNZ NEXT
HLT
```

Starting Address : 0000

Comments :

User Data Grid

Address	Data
4200	11
4201	22
4202	33
4203	AA
4204	BB
4205	CC
4206	DD
4207	EE

Hex Code Grid

Address	Data
0000	01
0001	00
0002	42
0003	11
0004	00
0005	43
0006	26
0007	07
0008	0A
0009	12
000A	03
000B	13
000C	25
000D	C2
000E	08
000F	00
0010	76

1.Data Transfer Group :

MOV	MVI	LDA	STA
LHLD	SHLD	LXI	XCHG
LDAX	STAX	IN	OUT

2.Arithmatic Group :

ADD	ADI	ADC	ACI	DAA
SUB	SUI	SBB	SBI	DAD
INR	DCR	INX	DCX	

3.Logical Group :

ANA	ANI	XRA	XRI	CMA
ORA	ORI	CMP	CPI	CMC
RLC	RRC	RAL	RAR	STC

4.Branch and Machine Control Group

JMP	J_	CALL	C_	
RET	R_	EI	DI	PUSH
PCHL	RST	RIM	SIM	POP
XTHL	SPHL	NOP	HLT	

Auto Insert | Insert | Cancel

Instruction Information :

LoaD H and L register direct

Ln 4, Col 4

TStates : 38 | MCycles : 12 | Time of Execution : 38 µs

Flags : S Z AC P CY

General Regs.:	A	B	C	D	E	H	L	M	SP	PC
	0A	00	00	00	00	00	00	3A	0000	0000

```
LDA 4200h
ANI 0Fh
STA 4201h
HLT
```

Starting Address : 0000

Comments :

User Data Grid

Address	Data
4200	4A
4201	0A
0000	3A
0000	3A
0000	3A
0000	3A
0000	3A
0000	3A

Hex Code Grid

Address	Data
0000	3A
0001	00
0002	42
0003	E6
0004	0F
0005	32
0006	01
0007	42
0008	76

1.Data Transfer Group :

MOV	MVI	LDA	STA
LHLD	SHLD	LXI	XCHG
LDAX	STAX	IN	OUT

2.Arithmatic Group :

ADD	ADI	ADC	ACI	DAA
SUB	SUI	SBB	SBI	DAD
INR	DCR	INX	DCX	

3.Logical Group :

ANA	ANI	XRA	XRI	CMA
ORA	ORI	CMP	CPI	CMC
RLC	RRC	RAL	RAR	STC

4.Branch and Machine Control Group

JMP	J_	CALL	C_	
RET	R_	EI	DI	PUSH
PCHL	RST	RIM	SIM	POP
XTHL	SPHL	NOP	HLT	

Auto Insert | Insert | Cancel

Instruction Information :

MoVe Immediate 8-bit

Ln 6, Col 4

TStates : 52 | MCycles : 16 | Time of Execution : 52 µs
Flags : S Z AC P CY

General Regs.:	A	B	C	D	E	H	L	M	SP	PC
	55	00	00	00	00	00	00	3A	0000	0000

```
LDA 4200h
CMA
STA 4201h
INR A
STA 4202h
HLT
```

Starting Address : 0000

Comments :

User Data Grid

Address	Data
4200	AB
4201	54
4202	55
0000	3A
0000	3A
0000	3A
0000	3A
0000	3A

Hex Code Grid

Address	Data
0000	3A
0001	00
0002	42
0003	2F
0004	32
0005	01
0006	42
0007	3C
0008	32
0009	02
000A	42
000B	76

1.Data Transfer Group :

MOV	MVI	LDA	STA
LHLD	SHLD	LXI	XCHG
LDAX	STAX	IN	OUT

2.Arithmatic Group :

ADD	ADI	ADC	ACI	DAA
SUB	SUI	SBB	SBI	DAD
INR	DCR	INX	DCX	

3.Logical Group :

ANA	ANI	XRA	XRI	CMA
ORA	ORI	CMP	CPI	CMC
RLC	RRC	RAL	RAR	STC

4.Branch and Machine Control Group

JMP	J_	CALL	C_	
RET	R_	EI	DI	PUSH
PCHL	RST	RIM	SIM	POP
XTHL	SPHL	NOP	HLT	

Auto Insert | Insert | Cancel

Instruction Information :

MoVe Immediate 8-bit